Consolidated Findings: Simple Policies, Information, and the Room for RL in the Pharmaceutical Supply-Chain Simulator

One running document consolidating the routing study, the routing-repaired ("understanding") study, and the value-decomposition study (Phases 1–2, distributor-seat test, Part C). Individual reports remain archived as the audit trail.

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Setting and scope

Simulated pharmaceutical supply chain from the predecessor thesis. The question: *can simple, transparent rules match the thesis's RL agent and beat base stock — and if not, where does the remaining value for learning live?* This document consolidates all three experiment campaigns, claims-first; superseded numbers are listed in Section 4 and should not be cited.

Table 1: Experimental setup (the thesis configuration; everything else varies by study and is stated where used).

Network	2 manufacturers → 2 distributors → 2 health centers; each HC dual-sourced
Demand	non-urgent $\mathcal{N}(120, 5)$ per health center per period; urgent 0 or 20
Costs	holding 1/unit, backlog 10/unit, revenue 10/unit
Disruption	one manufacturer at **5% capacity for 48 of 300 periods** (periods 110–157)
Capacity	400 units/period per manufacturer; normal load ≈120, so the survivor has 3.3× headroom
Baseline	base stock (order-up-to-target) at every echelon
Control seat	the RL agent (and our rules) control the distributors: one order quantity + the allocation split across health centers

Term	**Meaning**
thesis world / routing-repaired	The simulator as the thesis shipped it (one health center hard-wired to a 50/50 supplier split — the "captive" customer) vs. with our repaired health-center routing layer.
disruption cost	System-wide holding + backlog cost from onset to episode end.
moderate / severe scarcity	The surviving plant is *also* cut, to 50% / 30%, so total supply < total demand (the thesis scenario leaves it unconstrained — "slack").
urgent0 / urgent20	Demand configs: all demand deferrable vs. 20 urgent patients per health center per period, lost if unmet.
premium / coverage	Extra normal-times cost a policy pays vs. disruption damage it removes.
reporting seeds	20 held-out demand scenarios (11–30), never used for tuning; all headline numbers.
the compound	The named combination policy of the section at hand, e.g. the distributor compound: serve-the-captive × stop ordering from the dead factory × standing buffer.

1 Scorecard: the claims at a glance

One line per load-bearing claim; all numbers audited on held-out seeds. Full-sentence versions and per-claim evidence live in the hypothesis cards and archived reports; superseded numbers are in Section 4.

Claim	Key numbers	Status
The disruption pile-up is mostly a routing artifact, not supply physics	67–91% of cost removed by HC routing rules	established
Sharing the upstream machine-state signal = perfect onset knowledge: a two-line reroute matches the oracle exactly	cost 1.06M → 295k; lost 445 → 203	established
Downstream detection is blind for ∼6 periods (the failed distributor ships from its own stock)	∼0.9M accrues behind the mask	established
True foresight adds a small real margin (buffer must sit at the surviving chain)	≈54k (∼5%)	corrected
Simple distributor rules recover about HALF of the RL's claimed advantage	**49–51% vs claimed 89% (unreplicated)**	**open gap**
Demand-shaping through allocation is real and direction-sensitive	shed +21.8%; wrong direction −14.3%	established
Under scarcity, buffer LOCATION and demand-aware SIZING dominate; the compound owns the frontier	−79% at fill ≈1.0; lost 505 → 41	established
Lever-flip map: no action → routing → compound → buffer-only; robust to recurring disruptions	severe cell: buffer at disrupted chain, −30%	established
Every adaptive "smart" rule tested backfired (detection, five allocation rules, two redirect caps)	+9% to +56% worse than simpler alternatives	established
Gated rules cost nothing in normal times; standing buffers pay in full	no-disruption profit: −0.5% vs −34%	established
Our repaired DRL trains stably but loses to base stock (no working RL comparator)	system profit −4.06M vs −2.35M	established

2 The four findings that matter

F1 — If the distributors can see the manufacturers' machine status, a two-line rule is as good as perfectly predicting the disruption — and without that visibility, no algorithm of any sophistication can react in time. The evidence: at thesis severity the surviving manufacturer has 3.3× headroom and ramps its production with the orders it receives; a two-line reroute triggered by the shared upstream machine-state signal exactly reproduces the perfect-onset oracle (bit-identical trajectories). Without that sharing, no downstream policy — learned or not — can act in time, because the failed distributor's own stock masks the failure from every downstream observer for about six periods. *Meaning: the thesis's information-sharing narrative is right, but its value is captured by a trivial rule, not by a learner.*

F2 — A three-parameter rule recovers about half of the improvement the thesis's RL agent claimed over base stock; whether the other half is real cannot be judged until the thesis's baseline configuration is found. The evidence: from the distributor's seat, in the thesis's own world and on the thesis's own metric, the compound (serve-the-captive routing × a taper that stops ordering from the dead factory × a standing buffer) removes 49–51% of base stock's loss — against the RL's claimed 89% from a thesis table we cannot replicate (our own corrected-pipeline DRL agent *lost* to base stock). *Meaning: simple rules beat base stock decisively; whether the RL's*

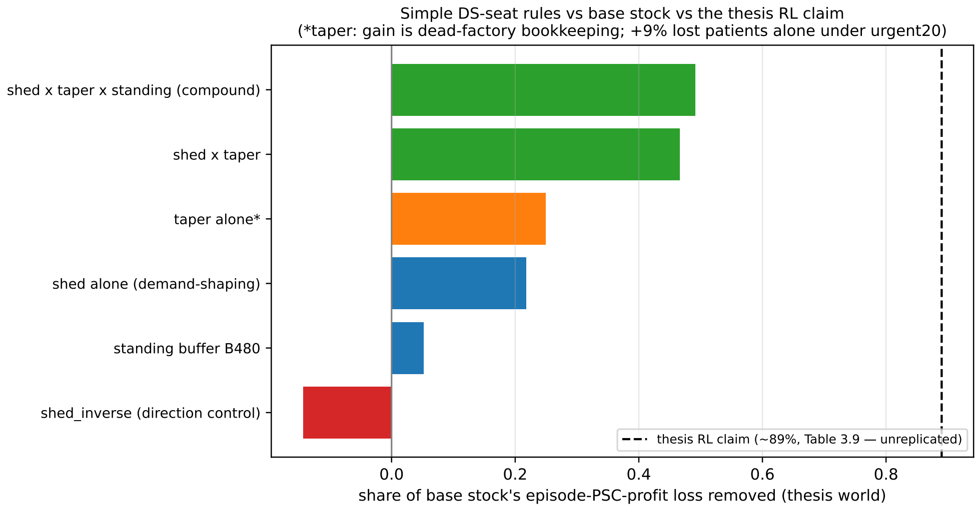

Figure 1: F2 in one picture: simple distributor rules vs the RL claim, thesis world, audited reporting seeds.

remaining margin is real is now a question about the thesis baseline configuration — the single most valuable item to obtain.

Equity of the shed rule: on episode totals both health centers are better off under the shed compound (with urgent patients present — the urgent20 configuration — lost patients fall at the trust-routed health center 804→713 and at the captive health center 964→862; during-disruption fill rises for both), but the transition has a real localized cost: the deliberately starved health center's worst 5-period fill deepens from 0.614 (base stock) to ≈0.50 before the reroute completes. The rule is Pareto-improving on totals with a measurably deeper short trough for the starved customer — the honest ethics framing for any deployment.

F3 — When supply is genuinely short, what matters is where reserve stock sits and how big it is — both computable in advance from observable quantities; reacting cleverly during the outage does not help. The evidence: under scarcity the cost–service frontier is owned by a single composition (buffer + reroute + taper); the buffer's optimal *location* flips to the disrupted chain once the surviving chain is too weak to carry the load; and demand-aware *sizing* — buffer = (demand − surviving deliverable) × duration — cuts lost patients from 505 to 41. *Meaning: the two deployable decisions that matter are where stock sits and how it is sized — both computable from observables, no learning required.*

F4 — Every adaptive "smart" rule we tested made the system worse; simple static structure won every time. The evidence: across three studies, every reactive rule backfired — delivery-rate detection, backlog-priority allocation, serve-the-captive (in the routing-repaired world), rotating priority, priority-with-floor, and both severity-aware redirect caps. The mechanism is structural: the trust feedback loop rewards consistency, and the surviving manufacturer's production is driven by the orders it receives, so clever order withholding starves the very supply it is trying to protect. *Meaning: for the eventual RL comparison, the honest prior is that adaptivity must overcome a measured pattern of adaptive rules underperforming static structure.*

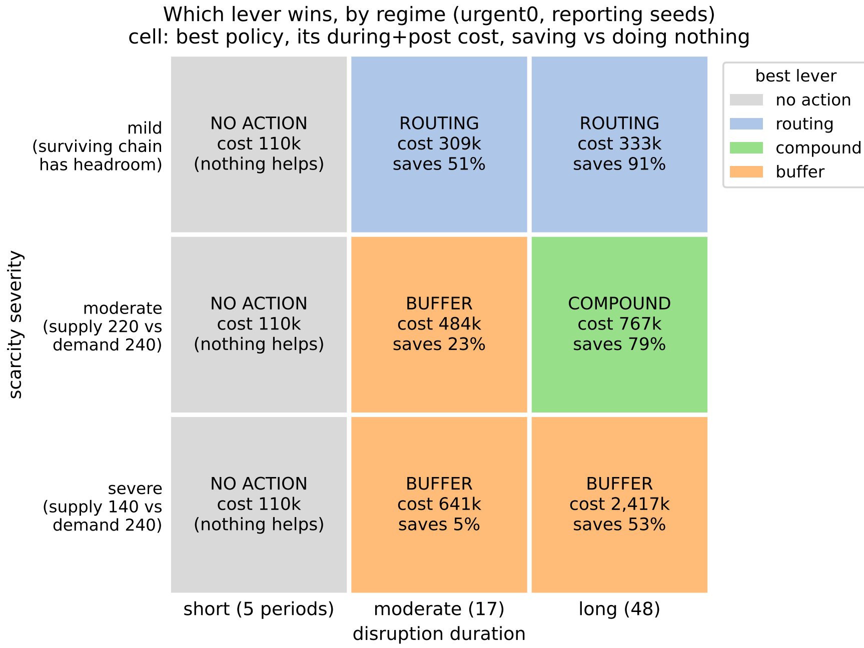

Figure 2: F3 in one picture: the lever-flip map over the severity × duration plane. Each cell's color indicates the winning lever (no-action → routing → compound → buffer-only); the severe-scarcity cell is corrected to a disrupted-located buffer. Cell color encodes WHICH lever wins; cell text gives that policy's cost and its saving versus doing nothing. The recurring-disruption robustness check has since been run and PASSED: with two events per episode no cell changes its winner, so the map stands.

3 Phase-by-phase detail

3.1 Background: the DRL arc (why there is no working RL comparator)

The reward audit found reward components 1 and 4 carrying a thesis-level algebraic defect, component 5 a code defect, and components 2/3/6 saturated as configured; repairs un-stuck all six components. Training instability traced to the adaptive exploration rule (a multiplicative update that runs away to its cap); a linear schedule yields a stably-training agent — which then *loses* to base stock on system profit (−4.06M vs −2.35M, driven by manufacturer queue cost). The thesis's headline RL result (its Table 3.9) is therefore treated as a claim, not a reproduced result, pending the author's configuration.

3.2 Routing study (thesis severity)

The captive health center's hard-wired equal split, plus stranded-order suppression (orders stuck in a dead supplier's queue are never re-issued), lock ∼2/3 of the disruption cost into the routing layer: trust-split −26%, sharpened trust + stale-order write-off −67% (fill 1.000), oracle onset reroute −91%; lost patients 1,768 → 203. Found en route: the normal demand model hard-reseeds the global random number generator (all historical multi-seed normal-demand results were a single demand path) — fixed in our runners, gate-guarded.

3.3 Understanding study (routing-repaired world)

Allocation bounded at $\leq$12% via the maximum over the enumerated rule family (static priority; fairness cost $\pm$0.023 fill); all dynamic rules backfire through the trust loop. The residual disruption cost in the routing-repaired world decomposes into 41% dead-factory queue bookkeeping (dual cost accounting was introduced for exactly this), the $\sim$10-period redirection transient, and a write-off glut later removed as a side effect of instant rerouting.

3.4 Value decomposition, Phase 1 (slack regime)

Info-sharing reroute $\equiv$ oracle (bit-identical, audited); explicit delivery-rate detection worse than the built-in trust mechanism (stock-cover masking $\sim$0.9M); deployable compound 273k. Verification rebuilt the oracle as a true superset and corrected two claims: foresight is worth $\sim$54k with the buffer at the SURVIVING chain, and slack-regime buffers are negative only at the disrupted location (+4–7% at the survivor).

3.5 Value decomposition, Phase 2 (scarcity regimes)

With the surviving manufacturer also cut (to 50% / 30%), routing still reduces cost ($-$26%/$-$50%) but cannot create product (lost patients $\sim$flat across the entire ladder); the compound reaches $-$79% at fill 0.99–1.00; static priority flips harmful (+23%); the severity$\times$duration map has two boundaries (stock-cover duration; surviving-chain headroom), with sharp rerouting the WORST measured policy at severe scarcity.

3.6 Distributor-seat study (thesis world) and Part C

Full-grid ladder over the distributor's complete decision space (everything else is dead by topology). Results in F2/F4 and the claims tables. Part C: demand-aware buffer sizing (the robust knob), the severe-scarcity buffer-location correction, and the redirect-cap refutation with its mechanism (orders are the surviving supplier's production signal).

4 Superseded claims (do not cite)

- The $-$468k / $-$654k value split — measured on the broken routing layer; retracted.
- "Permanent insurance captures 42%" — broken-routing artifact; in the routing-repaired slack-regime world, disrupted-located buffers are zero-to-negative (surviving-located: +4–7%).
- "Foresight adds nothing beyond the shared signal" — artifact of a mis-located ceiling buffer; corrected to $\sim$54k ($\sim$5%).
- "Detection captures $\approx$0" — true only for detect-then-buffer; rerouting responses carry most of the onset value, and the upstream-signal trigger captures it fully.
- "Allocation $\leq$12% is the whole channel" — specific to the routing-repaired world; in the thesis world the demand-shaping channel is worth +21.8% (shed) and the compound 49%.
- "Fill 1.000" for the 960-unit-buffer scarcity compound — display-rounding; 0.990 (the 1,440-unit buffer reaches 0.9999).

- The frozen per-severity buffer grids — replaced by demand-aware sizing.

5 Rigor record (one paragraph)

All experiments ran on a gated harness (bit-exact baseline reproduction on both library and CLI paths, determinism, demand conservation, cross-seed variance), with pre-registered tuning/reporting seed splits (1–10 / 11–30), dual cost accounting, and both fairness baselines carried. Every phase was audited by an independent agent re-deriving headline numbers from raw CSVs (12/12, 11/12 — the exception a display-rounding over-claim, corrected — 16/16, 13/13, and 18/18 for the robustness checks). Three contaminated or buggy batches were caught by gates or diagnostics (CLI-default contamination; a bootstrap deadlock; both logged) and discarded without contaminating conclusions. Simulator defects found and guarded: the normal demand model's global RNG reseed, a dead trust-delta config constant, the silent discard of orders ≤ 1 unit, and the missing on-order timeout that produces stranded-pipeline suppression.

6 Next steps

1. **Obtain the thesis Table 3.9 configuration (incl. the base-stock baseline)** — decides the interpretation of F2's 49%-vs-89% gap; everything else is measured.
2. *(Done since the last draft:)* the recurring-disruption robustness check PASSED — the map's winners are unchanged and the compound amortizes best; coverage does not collapse in normal times (gated rules pay $\approx$zero premium); the shed rule's equity profile is characterized. The map can be published as-is.
3. Ramped-onset regime as the second paper (the one place detection could earn value).
4. When corrected DRL code arrives: evaluate against the regime-appropriate simple compounds (thesis world: the distributor-seat compound; slack regime: 273k; moderate scarcity: 767k–1,726k with demand-aware buffer sizing), same seeds, same gates, both accounting conventions.
5. Distill the paper from this document: the three-lever value decomposition with the lever-flip map, the distributor-seat result as the RL-relevant centerpiece.